

PCA in R

The protein data set is a real-valued multivariate data set describing the average protein consumption by citizens of 25 European countries.

For each country, there are ten columns. The first eight correspond to the different types of proteins. The last one corresponds to the total value of the average values of proteins.

The data have been downloaded from

<https://raw.githubusercontent.com/jgscott/STA380/master/data/protein.csv>

#loading the data

```
protein_data <- read.csv("protein.csv")
```

```
str(protein_data)
```

#showing first 5 rows and some statistics

```
head(protein_data)
```

```
summary(protein_data)
```

#Checking for missing values

```
colSums(is.na(protein_data))
```

#Normalizing the data

```
data_normalized <- scale(numerical_data)
```

```
head(data_normalized)
```

#Correlation matrix

```
install.packages("corrplot")
```

```
library(corrplot)
```

```
corr_matrix <- cor(data_normalized[, 1:9])
```

```
corrplot(corr_matrix, method = "color", addCoef.col = "black")
```

#Applying PCA

```
data.pca <- princomp(data_normalized)
```

```
summary(data.pca)
```

#Plotting Scree Plot

```
install.packages("factoextra")
```

```
library(factoextra)
```

```
fviz_eig(data.pca, addlabels = TRUE)
```